Louay Ouhibi

Bioprocess Engineer

Tunis, Tunisia • Open to relocation • Available immediately
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Professional Profile

Freshly graduated Industrial Biology Engineer specializing in bioprocess optimization, biogas production, and fermentation. I combine process engineering fundamentals with Python-based modeling to optimize operating conditions, anticipate process instability, and turn experimental data into actionable process decisions. Seeking an engineering role in biogas, fermentation, or low-carbon bioprocess projects.

Education

National Institute of Applied Sciences and Technology – INSAT *2021–2026*
Engineering Degree in Industrial Biology – Three-year engineering program
Preparatory Cycle in Applied Chemistry and Biology – two-year program
Relevant coursework: Process Engineering, Environmental Engineering, Fermentation, Thermodynamics, Fluid Mechanics, Heat and Mass Transfer, Industrial Microbiology, and Biochemistry

Professional Experience

Anaerobic Digestion and Biogas Process Optimization *May–August 2026*
End-of-Study Internship – Memorial University of Newfoundland, Canada

- Built a technical anaerobic digestion database covering feedstocks, reactor configurations, operating conditions, methane yield, and inhibition indicators.
- Developed decision-support models for methane-yield prediction and inhibition-risk screening, achieving a test R^2 of **0.909** for methane yield, **91.2% accuracy** for inhibition-risk detection, and **92.7% accuracy** for inhibition-type classification, and identified major inhibition pathways (ammonia, LCFA, salinity, VFA accumulation).
- Developed **AnaeroVision**, a user-oriented interface integrating methane-yield estimation, inhibition screening, and pathway diagnosis into one workflow for early process diagnosis.

Production Process Optimization – Pharmaceutical Industry *June–July 2025*
Industrial Intern – ADWYA Pharmaceutical, Alpha Production Unit

- Supported continuous-improvement activities aimed at reducing production time and improving process efficiency.

Organic Waste Valorization through Vermicomposting *July 2024*
Environmental Biotechnology Intern – International Centre for Environmental Technologies of Tunis

- Assessed the vermicomposting valorization potential of organic waste and evaluated the culture conditions affecting worm development and organic-matter conversion.

Selected Engineering Projects

BioPredict – Single-Cell Protein Performance Prediction *March 2026*
Independent Machine Learning and Bioprocess Engineering Project

- Integrated biological, fermentation, cost, and environmental data for six single-cell protein strains to assess process productivity and techno-environmental performance.
- Developed and validated regression models, with Random Forest achieving 5-fold cross-validation R^2 values of **0.976** for protein productivity, **0.832** for OPEX, and **0.805** for GHG emissions.
- Built **BioPredict**, a decision-support tool for strain screening, performance benchmarking, key process-driver analysis, and viability assessment.

Microalgae Cultivation in a Photobioreactor for Wastewater Treatment *May 2025*
Team Engineering Project – INSAT

- Studied the cultivation of *Chlorella* sp. in a photobioreactor using industrial wastewater and evaluated the influence of nutrients, temperature, pH, light, and aeration on biomass growth.
- Developed a predictive model achieving an R^2 of **0.9607** for microalgae-growth prediction from environmental and operating parameters.
- Built a Python interface for real-time growth prediction and process-condition assessment in wastewater bioremediation.

Technical Skills

Process Engineering: Process optimization, operating-condition analysis, reactor-performance assessment, fermentation
Data Analysis: Statistical analysis, regression and classification, model evaluation, data-driven decision support
Software & Tools: Python, Scikit-learn, R, Microsoft Excel, Word, PowerPoint, Linux, Galaxy

Extracurricular Activities

Media Manager – INSAT Theatre Day: communications, social media, and event coverage. *February 2025*
Internal Organization Manager – Lions Club Tunis INSAT: coordinated logistics and donation activities. *2023*

Languages

Arabic: Native • **French:** Fluent • **English:** Fluent • **German:** Beginner